

Analytical Distance Calculations between von Mises and Wrapped Cauchy Distributions

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June 16, 2026

1 Introduction

This document summarizes the mathematical details of the analytical distance calculations (KL divergence and 1-Wasserstein distance) between the von Mises distribution and the wrapped Cauchy distribution on the circle S^1 .

2 Distributions

Let P be the von Mises distribution with mean direction $\mu_P \in [-\pi, \pi)$ and concentration parameter $\kappa > 0$:

$$p(\theta) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu_P)} \quad (1)$$

where $I_n(\kappa)$ is the modified Bessel function of the first kind of order n .

Let Q be the wrapped Cauchy distribution with mean direction $\mu_Q \in [-\pi, \pi)$ and concentration parameter $\rho \in [0, 1)$:

$$q(\theta) = \frac{1 - \rho^2}{2\pi(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))} \quad (2)$$

3 Kullback-Leibler Divergence

3.1 Divergence from von Mises to Wrapped Cauchy

The KL divergence $D_{\text{KL}}(P \parallel Q)$ is defined as:

$$D_{\text{KL}}(P \parallel Q) = \int_0^{2\pi} p(\theta) \log \left(\frac{p(\theta)}{q(\theta)} \right) d\theta \quad (3)$$

Using the Fourier series expansion:

$$\log(1 + \rho^2 - 2\rho \cos x) = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx) \quad (4)$$

and the trigonometric moments of the von Mises distribution:

$$E_P[\cos(n(\theta - \mu_Q))] = \frac{I_n(\kappa)}{I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \quad (5)$$

we obtain the fast-converging series representation:

$$D_{\text{KL}}(P \parallel Q) = \kappa \frac{I_1(\kappa)}{I_0(\kappa)} - \log I_0(\kappa) - \log(1 - \rho^2) - 2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \quad (6)$$

3.2 Divergence from Wrapped Cauchy to von Mises

The KL divergence $D_{\text{KL}}(Q \parallel P)$ can be calculated in closed-form. Using $E_Q[\cos(\theta - \mu_P)] = \rho \cos(\mu_Q - \mu_P)$ and $E_Q[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] = 2 \log(1 - \rho^2)$, we get:

$$D_{\text{KL}}(Q \parallel P) = \log I_0(\kappa) - \log(1 - \rho^2) - \kappa \rho \cos(\mu_Q - \mu_P) \quad (7)$$

4 1-Wasserstein Distance (Aligned Means)

When the mean directions are aligned ($\mu_P = \mu_Q$) and one distribution is more concentrated than the other, the 1-Wasserstein distance $W_1(P, Q)$ on the circle reduces to the difference of their mean absolute deviations on $[-\pi, \pi]$:

$$W_1(P, Q) = |E_P[|\theta|] - E_Q[|\theta|]| \quad (8)$$

The mean absolute deviation of each symmetric distribution is given by:

$$E_P[|\theta|] = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{I_{2k+1}(\kappa)}{(2k+1)^2 I_0(\kappa)} \quad (9)$$

$$E_Q[|\theta|] = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\rho^{2k+1}}{(2k+1)^2} \quad (10)$$

Thus, the 1-Wasserstein distance is:

$$W_1(P, Q) = \frac{4}{\pi} \left| \sum_{k=0}^{\infty} \frac{\rho^{2k+1} - \frac{I_{2k+1}(\kappa)}{I_0(\kappa)}}{(2k+1)^2} \right| \quad (11)$$